

MNC Logistics Report — User Guide

This guide walks you through configuring and running the MNC Logistics Report workflow, which processes balloon landing, airstrip operations, airstrip maintenance, and airline complaint events from EarthRanger to produce tabular CSV reports and an interactive dashboard.

Overview

The workflow delivers, for each run:

- **balloon_landing_summary_table.csv** — passenger records grouped by balloon company and lodge
 - **airstrip_operations_summary_table.csv** — total client counts per camp/lodge pivoted by arrival and departure direction
 - **airstrip_maintenance_summary_table.csv** — dated log of airstrip maintenance activities
 - **A logistics dashboard** with an interactive, sortable/filterable table widget for each of the three summary tables above
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Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with `balloon_landing`, `airstrip_operations`, `airstrip_maintenance`, and `airline_complaint` events recorded for the analysis period
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Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/mnc_logistics_report.git
```

Then click **Add Template**.

Step 2 — Configure the EarthRanger Connection

Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

Field	Description
Data Source Name	A label to identify this connection (e.g. <code>Mara North Conservancy</code>)
EarthRanger URL	Your instance URL (e.g. <code>your-site.pandas.org</code>)
EarthRanger Username	Your EarthRanger username
EarthRanger Password	Your EarthRanger password

Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **logistics_report**. Click the card to open the workflow configuration form.

Step 4 — Configure Workflow Details, Time Range, and EarthRanger Connection

The configuration form has three sections on a single page.

Set Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. reporting month or site)

Time Range

Field	Description
Timezone	Select the local timezone (e.g. <code>Africa/Nairobi UTC+03:00</code>)
Since	Start date and time — all logistics events from this point are fetched
Until	End date and time of the analysis window

Connect to EarthRanger

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown (e.g. `Mara North Conservancy`).

Once all three sections are filled, click **Submit**.

Running the Workflow

Once submitted, the runner will:

1. Fetch `balloon_landing` , `airstrip_operations` , `airstrip_maintenance` , and `airline_complaint` events for the analysis period from EarthRanger as point geometries; extract the date from each event's timestamp; add a temporal index.
2. Filter `balloon_landing` events; process and flatten event details (mapping field IDs to display titles); drop the `event_details__` prefix; retain and rename relevant columns

- (`balloon_company` , `where_are_clients_staying` , `no_of_passengers`); clean bracket characters; rename columns to display-friendly headers (`Date` , `Balloon Company` , `Where Are Clients Staying` , `No Of Passengers`); save as `balloon_landing_summary_table.csv` and render as the **Balloon Landing Summary** dashboard widget.
3. Filter `airstrip_operations` events; process and flatten event details; drop the `event_details__` prefix; rename columns (`airline` , `arrival_departure` , `attendant` , `camp_lodge` , `no_of_clients`); clean bracket characters; convert client counts to integer; compute totals per camp/lodge by direction; pivot by arrival/departure; convert pivoted values to integer; rename columns to display-friendly headers (`Camp Lodge` , `Arrival` , `Departure`); save as `airstrip_operations_summary_table.csv` and render as the **Airstrip Operations Summary** dashboard widget.
 4. Filter `airstrip_maintenance` events; process and flatten event details; drop the `event_details__` prefix; retain date and maintenance type columns; rename columns to display-friendly headers (`Date` , `Maintenance Type`); save as `airstrip_maintenance_summary_table.csv` and render as the **Airstrip Maintenance Summary** dashboard widget.
 5. Filter `airline_complaint` events; process and flatten event details; drop the `event_details__` prefix. This branch is not yet published to a summary table or dashboard widget.
 6. Assemble the **MNC Logistics Dashboard**, combining workflow details, time range, groupers, and the three table widgets above.
 7. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS` .

Steps are automatically skipped if their input data is empty or an upstream step was skipped, so a run with no events of a given type will simply omit that branch's output rather than failing.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/` .

File	Description
balloon_landing_summary_table.csv	Passenger records by balloon company and lodge (columns: Date , Balloon Company , Where Are Clients Staying , No Of Passengers)
airstrip_operations_summary_table.csv	Total clients per camp/lodge pivoted by arrival and departure (columns: Camp Lodge , Arrival , Departure)
airstrip_maintenance_summary_table.csv	Dated log of airstrip maintenance activity types (columns: Date , Maintenance Type)

Dashboard

The workflow run also produces the **MNC Logistics Dashboard**, viewable in the workflow runner, with one interactive table widget per summary table:

Widget	Source table
Balloon Landing Summary	balloon_landing_summary_table.csv
Airstrip Operations Summary	airstrip_operations_summary_table.csv
Airstrip Maintenance Summary	airstrip_maintenance_summary_table.csv

Each widget is sortable and filterable in place; downloading directly from the widget is disabled — use the corresponding CSV output file for that.